Data Analysis

Dataset

Super Market Dataset

Details

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In [8]:
```python
# importing all the modules

import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import seaborn as sn
```

In [7]:
```python
# Reading the data

df = pd.read_csv("data/SuperMarket Analysis.csv")

df.head()
```

Out[7]:

Invoice ID	Branch	City	Customer type	Gender	Product line	Unit pri
- -	Alex	Yangon	Member	Female	Health and beauty	.
- -	Giza	Naypyitaw	Normal	Female	Electronic accessories	.
- -	Alex	Yangon	Normal	Female	Home and lifestyle	.
- -	Alex	Yangon	Member	Female	Health and beauty	.
- -	Alex	Yangon	Member	Female	Sports and travel	.

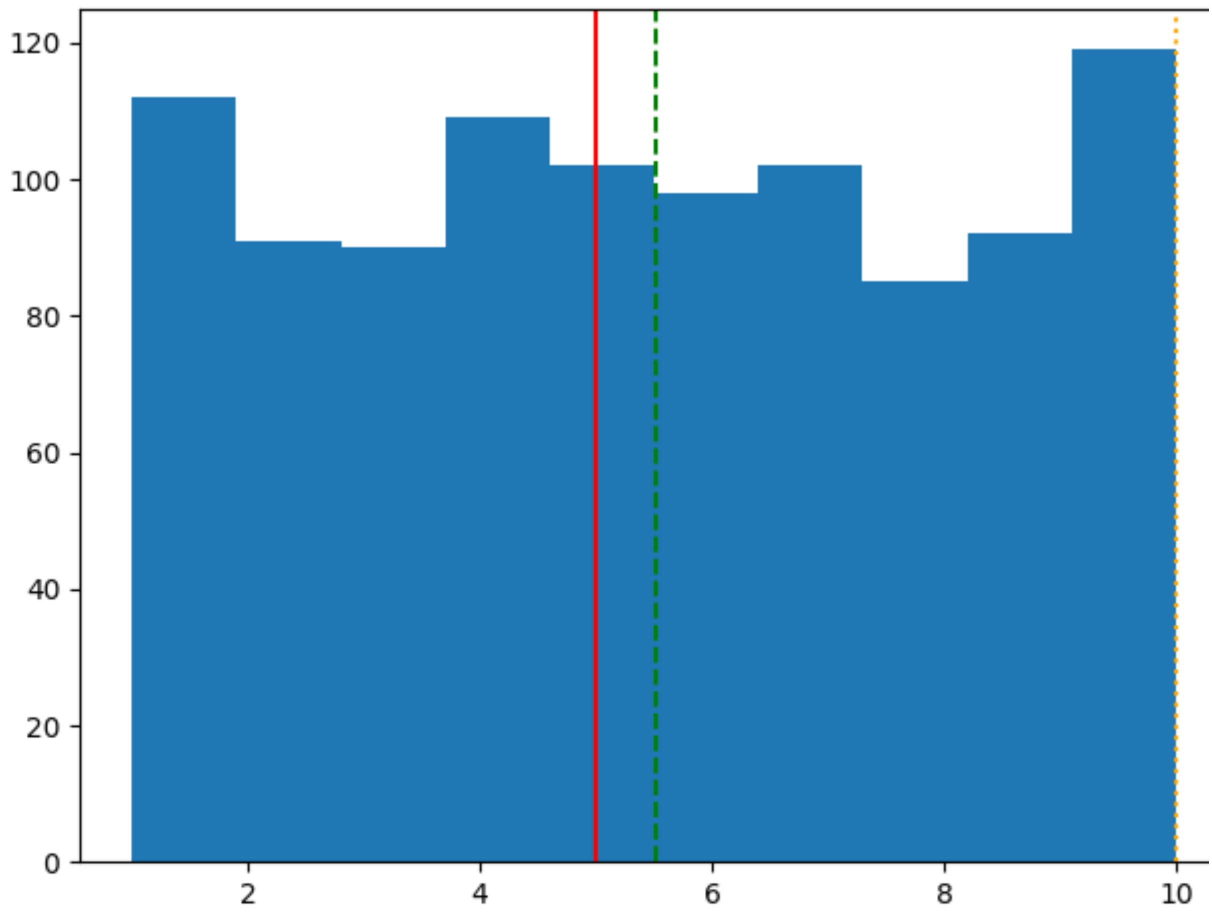
In [34]:
```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Invoice ID               1000 non-null   str
 1   Branch                   1000 non-null   str
 2   City                     1000 non-null   str
 3   Customer type            1000 non-null   str
 4   Gender                   1000 non-null   str
 5   Product line             1000 non-null   str
 6   Unit price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64
 8   Tax 5%                   1000 non-null   float64
 9   Sales                    1000 non-null   float64
 10  Date                     1000 non-null   str
 11  Time                     1000 non-null   str
 12  Payment                  1000 non-null   str
 13  cogs                     1000 non-null   float64
 14  gross margin percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
dtypes: float64(7), int64(1), str(9)
memory usage: 132.9 KB
```

In [58]:

```python
# Central tendency of data (Visual; Numeric)

def find_central_tendency(column_name):
    mean = df[column_name].mean()
    median = df[column_name].median()
    mode = df[column_name].mode()[0]
    print(f"The Mean: {mean}, Median: {median}, Mode: {mode}")

    # Diagram
    plt.hist(df[column_name],bins=10)

    # Central Line
    plt.axvline(mean, linestyle='--', label='Mean', color='green')
    plt.axvline(median, linestyle='-', label='Median', color='red')
    plt.axvline(mode, linestyle=':', label='Mode', color='orange')
    plt.tight_layout()
    plt.show()

find_central_tendency("Quantity")
```

```
The Mean: 5.51, Median: 5.0, Mode: 10
```

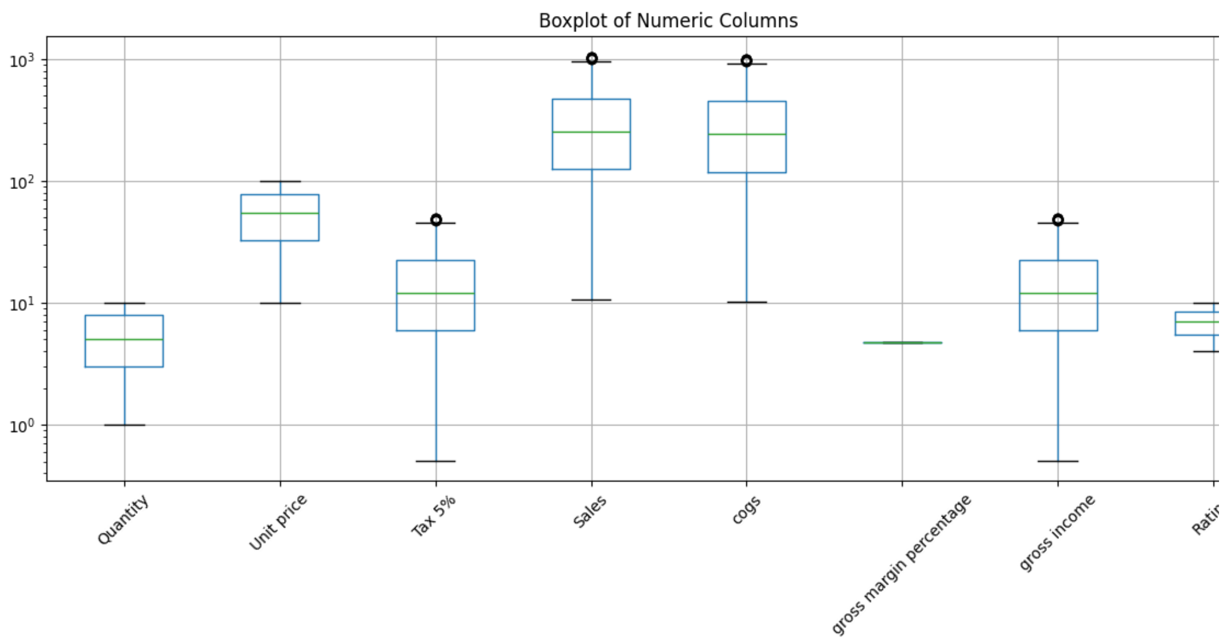


Outlier identification (Visual ; Numeric). Add outliers if the dataset does not have any.

In [54]:
```
for column in columns:

    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[
        (df[column] < lower) |
        (df[column] > upper)
    ]

    print(f"{column}: {len(outliers)} outliers")
```

```
Quantity: 0 outliers
Unit price: 0 outliers
Tax 5%: 9 outliers
Sales: 9 outliers
cogs: 9 outliers
gross margin percentage: 0 outliers
gross income: 9 outliers
Rating: 0 outliers
```

In [57]:
```
columns = [
    "Quantity",
    "Unit price",
    "Tax 5%",
    "Sales",
    "cogs",
    "gross margin percentage",
    "gross income",
    "Rating"
]

plt.figure(figsize=(12, 6))

df.boxplot(column=columns)
plt.yscale("log")

plt.title("Boxplot of Numeric Columns")
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

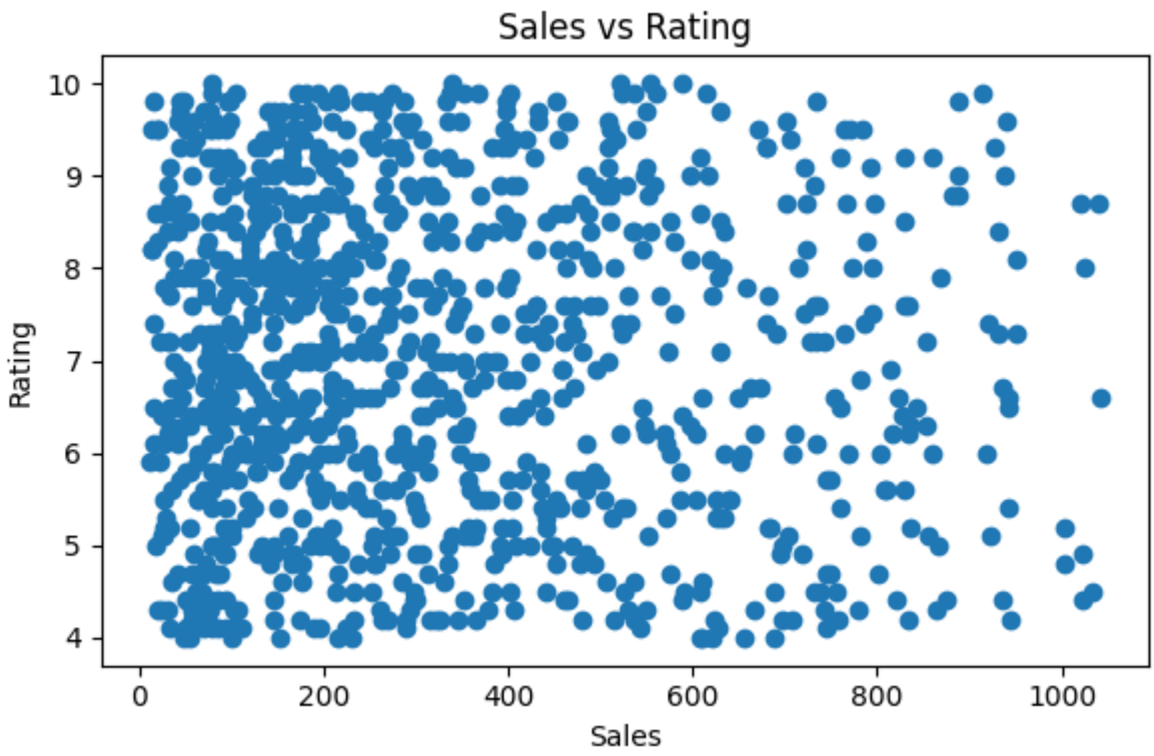


In this diagram there is outlires in **Tax** **%**, **Sales**, **Cogs**, and **Gross Income**.

Visualizing relationship between nume variables

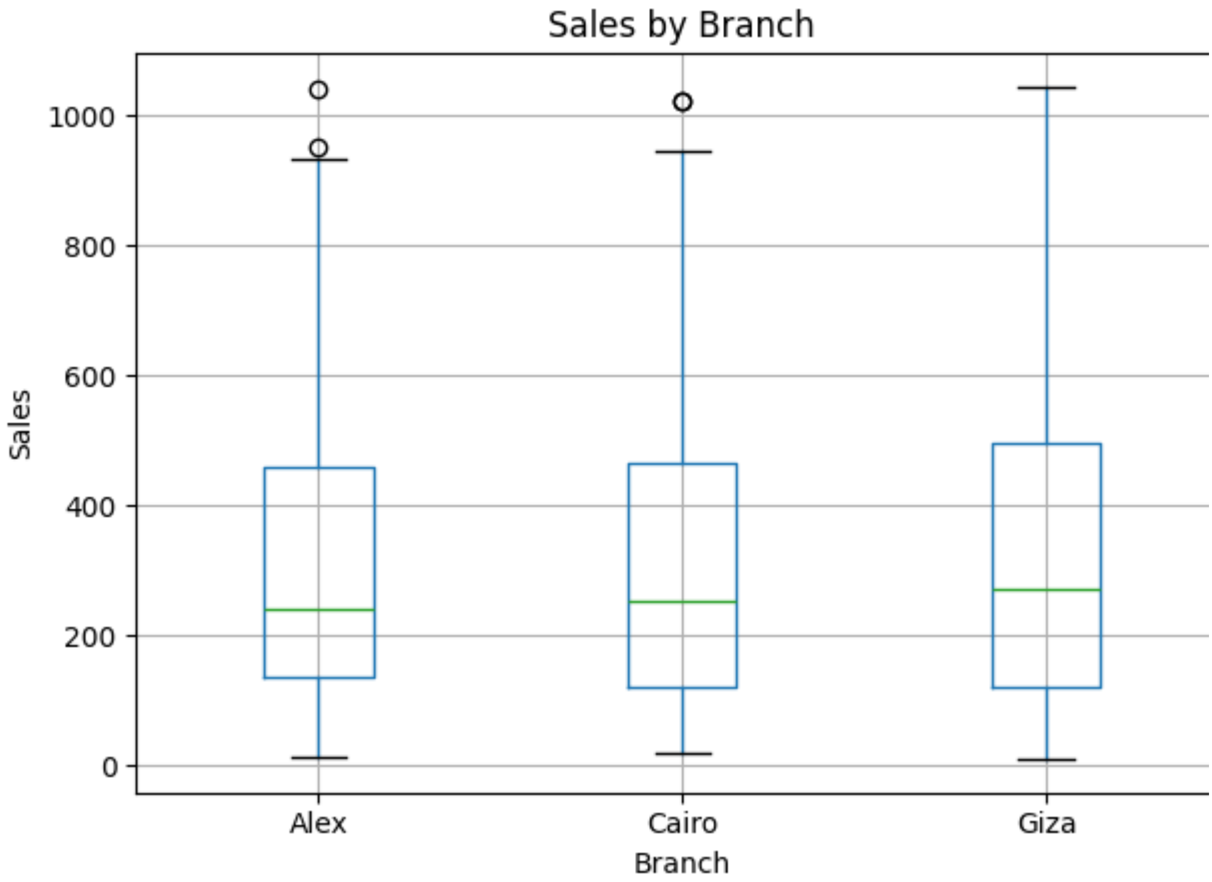
In [59]:
```
plt.figure(figsize=(6,4))

plt.scatter(df["Sales"], df["Rating"])

plt.xlabel("Sales")
plt.ylabel("Rating")

plt.title("Sales vs Rating")
```

```
plt.tight_layout()
plt.show()
```



Visualizing relationship between numeric and nominal variable

In [60]:
```
df.boxplot(column="Sales", by="Branch")

plt.title("Sales by Branch")
plt.suptitle("")  # removes extra automatic title

plt.xlabel("Branch")
plt.ylabel("Sales")

plt.tight_layout()
plt.show()
```

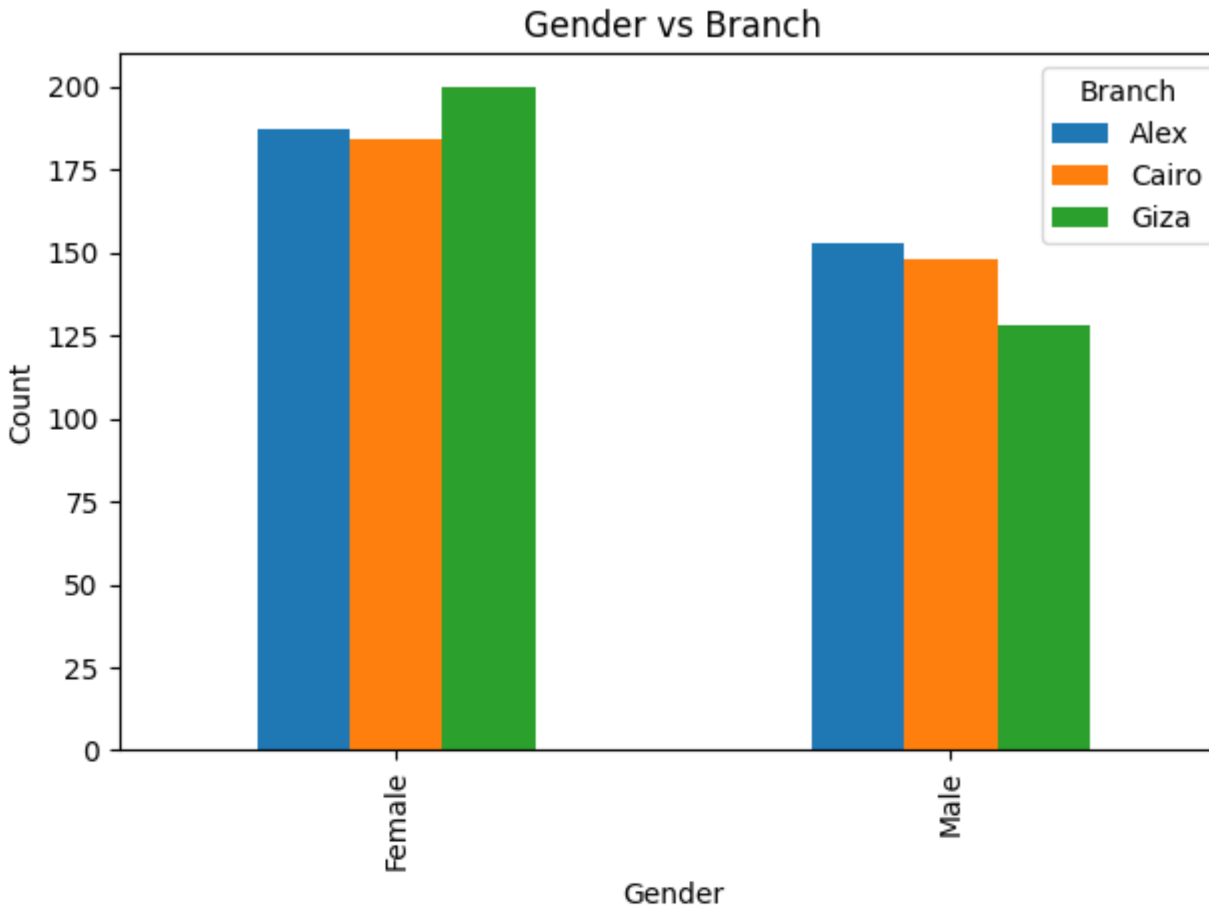


Visualizing relationship between two nominal variables.

In [52]:
```
cross_tab = pd.crosstab(df["Gender"], df["Branch"])

print(cross_tab)
```

```
Branch   Alex  Cairo  Giza
Gender
Female    187    184   200
Male      153    148   128
```

In [61]:
```
cross_tab.plot(kind="bar")

plt.xlabel("Gender")
plt.ylabel("Count")

plt.title("Gender vs Branch")

plt.tight_layout()
plt.show()
```


Gender vs Branch
200
175
150
125
100
75
50
25
0
Count
Branch
Alex
Cairo
Giza
Female
Male
Gender

In [ ]: